



The WMO Information System (WIS) 2.0 Framework for Data Driven AI Modelling

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ABSTRACT

WIS 2.0 is a powerful data source for development of AI weather models. It leverages the public internet, HTTP, and MQTT (Message Queuing Telemetry Transport) protocols. Data is accessible through well-documented REST (Representational State Transfer) APIs and OGC standards for automating data retrieval pipelines. This is ideal for AI models that need a constant feed of real-time data for predictions or for training on large, up-to-the-minute datasets. Data is provided in standard formats like BUFR, GRIB, and NetCDF, which are specifically designed for numerical weather prediction and are readily consumed by scientific Python libraries. Tools like “pywiscat”, “wis2downloader” and the API significantly lower the barrier to entry and provide a “plug and play” capability, allowing the developer to get started in hours rather than months. In a review publication [1] we reported in 2025 how WIS 2.0 will serve all users with free WMO core data. Here we try to look into current operational status of implementation world wide in general and India in particular for a research scientist. A researcher can find/bind with WIS2.0 in three steps – (1) find from Global Discovery Catalogue (GDC): The “library card catalog” for WIS 2.0. It’s an API endpoint where one can search for available datasets using metadata like location, time, or keywords (2) subscribe to Global Broker (GB): The “real-time news feed”. It uses an MQTT broker to publish notifications whenever new data is available. One subscribes to this feed to get alerted instantly. (3) bind with Global Cache (GC): The “download server” for core data. It provides direct HTTP(S) links to download the data files mentioned in the notifications from the Global Broker. As examples (1) we download initial conditions (GRIB analysis) through WIS2 downloader and run the pretrained AI model for 5 day forecast and (2) secondly, use historic data from archives to train, test and forecast with a time series. [1] Chakrabarti Dulal, Chug Sunny, Nath Sankar and Sai Krishnan K. C. , 2025, “Technological Evolution of Communication and Dissemination of Meteorological Data and Information”, Mausam, 76, 1 , 1117-1127

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